UMAMI DAO

Deconstructed Delta Minimized

GLP Vaults

May 2023

OVERVIEW

GLP Vaults

A Trustless DeFi strategy that **deconstructs GLP** into underlying assets to achieve **targeted exposure** and attractive **delta-minimized returns**

Deconstructs GLP into key crypto assets: **USDC, BTC, ETH, LINK, UNI**

Deconstruction of GLP enables **Internal Netting**, reducing fees and **increasing returns**

Based on the yield of GMX's GLP product, which has historically paid **20-50% APR in $ETH**

Depositors can purchase **insurance***

Audited in 2 rounds by Zokyo

* Insurance available on a limited quantity, based on the capacity of third-party services

DeFi Market Yield Overview

GLP Vaults are the indisputable choice for **maximizing APR on core crypto assets**

USDC, USDT, DAI

Compound: USDC **2.10%**
Compound: USDC **2.64%**
AAVE: USDT **2.17%**
Whole list

ETH

Lido: **5.00%**
CBETH: **4.41%**
Rocket Pool: **5.17%**
Whole list

BTC

Aave: **0.08%**
Compound: **0.02%**
Venus: **0.37%**
Whole list

LINK

Aave: **0.22%**
Bancor: **0.04%**
Staked Link: **5.29%**
Whole list

UNI

Compound: **0.36%**
Aave: **0.01%**
Whole list

Source: defillama.com, May 2023, Base APY, excluding GLP, excluding liquidity pools, no additional incentives; Sorted by TVL; networks: Ethereum, Arbitrum, Polygon, Optimism, Avalanche, BSC

WHAT IS

GLP Token

- **An index of assets** used for swaps and leverage trading, acting as the counterparty to trading on the GMX platform
- Holders **provide liquidity**, profiting when traders lose and vice versa
- Holders earn 70% of all platform fees **distributed in $ETH**
- **Holders are exposed to the price of underlying (and sometimes undesirable) assets**
- For detailed stats, visit **stats.gmx.io**

PROGRESS

V1 to V2 Improvements

GLP Deconstruction
Breaks down GLP into individual asset for optimal hedging mechanics

In-house Model Development
V2 model fully developed internally

Statistical Optimization
Includes transaction cost analysis and hyperparameter tuning

Pay off Matching
V2 matches linear payoffs with linear hedges

Internal Netting
Potential for zero-fee delta hedging between Vault assets

Slow Release Staking
Liquid and Boosted vault

PROGRESS

V1 to V2 Improvements

Correlation Hedging
In-house correlation model to generate hedges in illiquid low index composition assets

Multiple Hedging Venues
V2 vaults utilize two hedging venues with compatibility to add more

Hedging Composability
Flexibility to add multiple different hedging platforms

Trader PnL Hedging
Optionality Option to hedge GMX's trader PnL in the future

Vault Framework
Scalable, potentially applicable to other LP/index style tokens

INITIAL TESTING

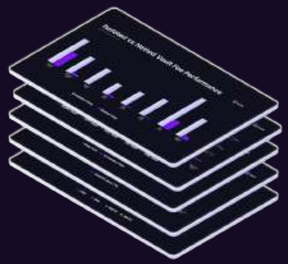
Backtesting Results

26.67
%APR

The Final Strategy, incorporating **Internal Netting**, demonstrated even stronger performance, reducing hedging costs and increasing TVL

11.80
%APR

The Baseline Strategy, i.e. **without Internal Netting**, for the USDC Vault still showed very positive returns



Our Litepaper and initial full backtesting results are available on our blog:

Backtesting Results

APRs reflect annualized performance during 8 mo. backtest period; positive-outliers are omitted for more conservative estimate.
Baseline Strategy excludes efficiencies from internal-netting.

BACKTESTING RESULTS

Isolated vs Netted Vault Fee Performance

The Final strategy reduced hedging costs by **65%-95%** with internal netting. Vault TVL in GLP was also increased by 11%.

PERFORMANCE IMPROVEMENT

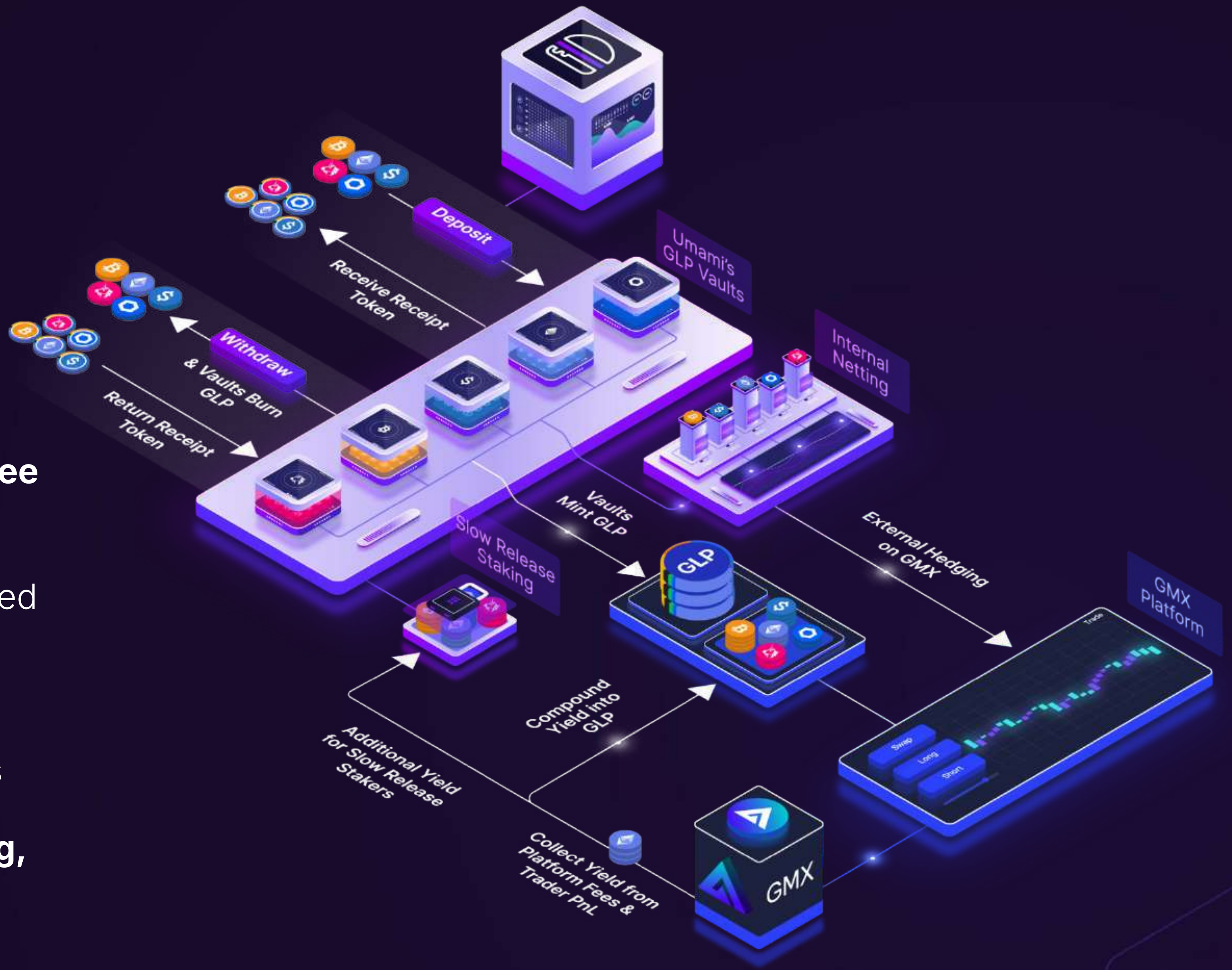
Slow Release Staking

- Liquid Vault for instant withdrawals and Boosted Vault for **time-locked** withdrawals or **Slow Release Staking**
- Boosters/Stakers get higher APY due to a slow-release withdrawal mechanism that improves overall vault performance
- 14-day wait for Boosted Vault withdrawal, but assets continue to yield during this time
- Deposits in Boosted Vault represented by **ERC-4626** tokens may have instant liquidity via Uni V3 Pool

GLP VAULTS

Additional Info

- Umami charges a **1.5% management fee** and a **20% performance fee**
- Yearly performance fee is targeted based on rebalance drawing of fees
- Vault deposits are represented by **ERC-4626** receipt implemented tokens
- Receipt ERC-4626 tokens allow **looping, borrowing** through other platforms



MULTIPLE SUSTAINABLE ADVANTAGES

Advantages Over Competition

- **More frequent rebalancing -** every 2 hours compared to the traditional 7 days. This enables us to achieve remarkably precise delta-tracking.
- **Internal Netting** permits a large percentage of our trades to be executed without any cost, thereby decreasing the overall hedging expenses. Additionally, it enhances **capital efficiency** by minimizing collateral requirements. This allows for **increased investment into GLP**, outpacing even the most experienced traders.
- We operate under realistic market conditions, contrary to competitors who attempted to replicate our internal netting method. **We DO externally hedge any excess deltas** above internal netting, and this allows us to scale the TVL according to our strategic needs.
- **Multiple liquidity sources** for hedging engineered into the vaults, guaranteeing not only **continuous availability of liquidity**, but also **the most cost-effective routing** for our external hedges at all times.
- **Lower/competitive fees**
- Option to boost yield by committing to a **Slow Release Staking**
- **More and closer partnerships** with protocols keen on utilizing our vault **ERC-4626 tokens** for diverse purposes, including providing **leverage, boosting liquidity, yield separation**, and more
- **Hedging assets 1:1** (ETH-ETH, BTC-BTC etc.) achieving greater precision in delta tracking and **reduce the tail risks**.

GLP Vaults TVL Growth Plan

Phase 1

Beta Release

- Vaults launch during a closed beta, during which ONLY Umami treasury assets can deposit into vaults.

Phase 2

Whitelist Launch

- Only the USDC Vault allows deposits
- To be eligible for depositing, you must hold $UMAMI (or mUMAMI, cmUMAMI)

Phase 3

Grow TVL

- DAOs (and similar on-chain institutions) are targeted for TVL commitments based on long term partnerships
- On-chain retail encouraged to deposit (especially in unbalanced vaults)

Phase 4

Maintain Cap

- A potential vault is available for hedging GLP's profit and loss (PnL)
- Potential additional vaults built on top of GLP Vaults

Develop on-chain strategies that deliver **high yield, hedge risk** and create a **sustainable** DeFi ecosystem

Careful integrated **risk analysis** allows undesirable exposure and uncorrelated risk hedging

Advanced DeFi strategies that capture high-return opportunities built upon **Real Yield***

Products are **designed for scale**, ensuring growing TVL creates a more-stable underlying system

*Real Yield = yield derived extrinsically from putting capital to work (eg. collecting fees) and not from internally minted assets (eg inflationary tokens)

UMAMI DAO

Home of Real Yield on Arbitrum

umami.finance

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